CodingChallenge6_IterationsandFunctions

GB

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Contents

Question1 **1**

Question 2. **1**

Question 3 **2**

Question 4 (Function Writing) **2**

Question 5 **3**

Question 6 (Loop) **3**

Qusetion 7 (dataframe) **4**

Question 8 (Github) **6**

Question1

1. Regarding reproducibility, what is the main point of writing your own functions and iterations?

Ans: The main purpose is to make analyses consistent, repeatable, and less error-prone. When you write your own functions and loops:

The same steps are applied exactly the same way every time

You avoid manual repetition, which reduces human error

Your workflow becomes transparent and easy to reuse

Others (or your future self) can understand and reproduce the analysis by running the same code

Question 2.

In your own words, describe:

(1) how to write a function in R Ans: A function in R is a reusable block of code designed to perform a specific task.

- Basic syntax: We define a function by assigning it a name and using the function() keyword, with inputs(arguments) inside parentheses. -The actual operations are written inside curly braces{} -The function returns the final results either explicitly using a return statement or implicitly as the last evaluated expression.

Simply, we give the function inputs, it performs operations, and it oitputs a result.

(2) how to write a for loop in R. In your answer, be specific about: basic syntax, where the code is written, and how results are returned or stored. (Answer in plain text; no code required for this question.)

Ans: A for loop is used to repeat a set of operations over a sequence of values.

Basic syntax: It starts with 'for', followed by a variable that takes on each value in a sequence.

Where code is written: The repeated operations are placed inside curly braces {}.

Storing results: Results are usually stored in a pre-created object (like a vector or list), which gets updated during each iteration.

In simple terms, a for loop runs the same code multiple times, changing the input each time and storing the results.

Question 3

```r
Cities <- read.csv(file.choose())
head(Cities)
```

```
##          city  city_ascii state_id state_name county_fips county_name     lat
## 1    New York    New York       NY   New York       36081      Queens 40.6943
## 2 Los Angeles Los Angeles       CA California        6037 Los Angeles 34.1141
## 3     Chicago     Chicago       IL   Illinois       17031        Cook 41.8375
## 4       Miami       Miami       FL    Florida       12086  Miami-Dade 25.7840
## 5     Houston     Houston       TX      Texas       48201      Harris 29.7860
## 6      Dallas      Dallas       TX      Texas       48113      Dallas 32.7935
##        long population density
## 1  -73.9249   18832416 10943.7
## 2 -118.4068   11885717  3165.8
## 3  -87.6866    8489066  4590.3
## 4  -80.2101    6113982  4791.1
## 5  -95.3885    6046392  1386.5
## 6  -96.7667    5843632  1477.2
```

Question 4 (Function Writing)

```r
Haversine<- function(lat1, lon1, lat2, lon2) {rad.lat1 <- lat1 * pi/180
  rad.lon1 <- lon1 * pi/180
  rad.lat2 <- lat2 * pi/180
  rad.lon2 <- lon2 * pi/180

  # Haversine formula
  delta_lat <- rad.lat2 - rad.lat1
  delta_lon <- rad.lon2 - rad.lon1
  a <- sin(delta_lat / 2)^2 + cos(rad.lat1) * cos(rad.lat2) * sin(delta_lon / 2)^2
  c <- 2 * asin(sqrt(a))

  # Earth's radius in kilometers
  earth_radius <- 6378137

  # Calculate the distance
  distance_km <- (earth_radius * c) / 1000

  return(distance_km)
}
colnames(Cities)
```

```
##  [1] "city"        "city_ascii"  "state_id"    "state_name"  "county_fips"
##  [6] "county_name" "lat"         "long"        "population"  "density"
```

Question 5

```r
auburn <- Cities$lat[Cities$city == "Auburn" ]
auburn2 <- Cities$long[Cities$city == "Auburn" ]

nyc <- Cities$lat[Cities$city == "New York"]
nyc2 <- Cities$long[Cities$city == "New York"]

# Calculate distance
 auburn_nyc_dist <- Haversine(auburn, auburn2, nyc, nyc2)

auburn_nyc_dist
```

```
## [1] 1367.854
```

Question 6 (Loop)

```r
for (i in 1:nrow(Cities)) {

  dist <- Haversine(auburn, auburn2,
                    Cities$lat[i],
                    Cities$long[i])
```

```r
  print(dist)
}
```

```
## [1] 1367.854
## [1] 3051.838
## [1] 1045.521
## [1] 916.4138
## [1] 993.0298
## [1] 1056.022
## [1] 1239.973
## [1] 162.5121
## [1] 1036.99
## [1] 1665.699
## [1] 2476.255
## [1] 1108.229
## [1] 3507.959
## [1] 3388.366
## [1] 2951.382
## [1] 1530.2
## [1] 591.1181
## [1] 1363.207
## [1] 1909.79
## [1] 1380.138
## [1] 2961.12
## [1] 2752.814
## [1] 1092.259
## [1] 796.7541
## [1] 3479.538
## [1] 1290.549
## [1] 3301.992
## [1] 1191.666
## [1] 608.2035
## [1] 2504.631
## [1] 3337.278
## [1] 800.1452
## [1] 1001.088
## [1] 732.5906
## [1] 1371.163
## [1] 1091.897
## [1] 1043.273
## [1] 851.3423
## [1] 1382.372
## [1] 0
```

Qusetion 7 (dataframe)

```r
distance_df <- NULL

for (i in 1:nrow(Cities)) {
    result_i <- data.frame(
      City1 = Cities$city[i],
```

```r
      City2 = "Auburn",
      Distance_km = Haversine(auburn, auburn2,
                              Cities$lat[i],
                              Cities$long[i]))

    distance_df <- rbind.data.frame(distance_df, result_i)
}
print(distance_df)
```

```
##            City1  City2 Distance_km
## 1       New York Auburn   1367.8540
## 2    Los Angeles Auburn   3051.8382
## 3        Chicago Auburn   1045.5213
## 4          Miami Auburn    916.4138
## 5        Houston Auburn    993.0298
## 6         Dallas Auburn   1056.0217
## 7   Philadelphia Auburn   1239.9732
## 8        Atlanta Auburn    162.5121
## 9     Washington Auburn   1036.9900
## 10        Boston Auburn   1665.6985
## 11       Phoenix Auburn   2476.2552
## 12       Detroit Auburn   1108.2288
## 13       Seattle Auburn   3507.9589
## 14 San Francisco Auburn   3388.3656
## 15     San Diego Auburn   2951.3816
## 16   Minneapolis Auburn   1530.2000
## 17         Tampa Auburn    591.1181
## 18      Brooklyn Auburn   1363.2072
## 19        Denver Auburn   1909.7897
## 20        Queens Auburn   1380.1382
## 21     Riverside Auburn   2961.1199
## 22     Las Vegas Auburn   2752.8142
## 23     Baltimore Auburn   1092.2595
## 24     St. Louis Auburn    796.7541
## 25      Portland Auburn   3479.5376
## 26   San Antonio Auburn   1290.5492
## 27    Sacramento Auburn   3301.9923
## 28        Austin Auburn   1191.6657
## 29       Orlando Auburn    608.2035
## 30      San Juan Auburn   2504.6312
## 31      San Jose Auburn   3337.2781
## 32  Indianapolis Auburn    800.1452
## 33    Pittsburgh Auburn   1001.0879
## 34    Cincinnati Auburn    732.5906
## 35     Manhattan Auburn   1371.1633
## 36   Kansas City Auburn   1091.8970
## 37     Cleveland Auburn   1043.2727
## 38      Columbus Auburn    851.3423
## 39         Bronx Auburn   1382.3721
## 40        Auburn Auburn      0.0000
```

Question 8 (Github)

CodingChallenge6